

Multivariate Analysis: Final Test Bank

Elementary Concentration and Correlation Inequalities

Conventions. $Z \sim N(0, 1)$ has CDF Φ ; $\mathbf{g} \sim N(\mathbf{0}, \mathbf{I}_n)$. For a square matrix, $|\mathbf{A}|$ is the determinant and $\mathbf{A} \succ 0$ means symmetric positive definite. var, cov are variance and covariance. The error function is $\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-u^2} du$. A *Rademacher* variable takes the values $+1$ and -1 with probability $\frac{1}{2}$ each (mean 0, variance 1); the symbol ε_i always denotes independent such variables.

Facts you may cite without proof.

- (F1) $1 + x \leq e^x$ for all $x \in \mathbb{R}$; Cauchy–Schwarz $\mathbf{E}[UV]^2 \leq \mathbf{E}U^2 \mathbf{E}V^2$; Jensen’s inequality.
- (F2) For $X \sim N(0, \sigma^2)$, $\mathbf{E}e^{\lambda X} = e^{\sigma^2 \lambda^2 / 2}$; $\int_{\mathbb{R}} e^{\mu u^2 - u^2 / 2} \frac{du}{\sqrt{2\pi}} = (1 - 2\mu)^{-1/2}$ for $\mu < \frac{1}{2}$; and $\mathbf{P}(|Z| \leq a) = 2\Phi(a) - 1 = \text{erf}(a/\sqrt{2})$.
- (F3) A density equals the inverse Fourier transform of its characteristic function, and for Gaussian-type integrands one may differentiate under the integral sign.
- (F4) Symmetric matrices are orthogonally diagonalizable.
- (F5) $\varphi(y) = (e^y - 1 - y)/y^2$ is nondecreasing on $\mathbb{R}$ (with $\varphi(0) = \frac{1}{2}$).
- (F6) Union bound: $\mathbf{P}(\bigcup_i A_i) \leq \sum_i \mathbf{P}(A_i)$.

Orientation. A *concentration inequality* bounds the probability that a random quantity strays from its mean. Problems 1–2 give the two ways to turn information about moments into a tail bound: Markov/Chebyshev (one

or two moments, *polynomial* decay) and the Chernoff method (the whole moment generating function, *exponential* decay). Problems 3–11 develop the Chernoff method into the standard tools for sums of independent variables — Hoeffding, the multiplicative Chernoff bound, Bennett–Bernstein — and adjoin a bound for maxima, a second-moment *lower* bound, and the χ^2 case. Problem 12 is the flagship application, the Johnson–Lindenstrauss lemma. Problems 13–15 change subject to the *positive correlation* of symmetric Gaussian events, ending with the bivariate Gaussian correlation inequality and a measurement of its slack.

Problem 1 (Markov, Chebyshev, and Cantelli). These three bounds are the bedrock of the subject: they assume almost nothing — no independence, no special distribution, only a moment or two — and everything later refines them. Let X have mean μ and variance $\sigma^2 < \infty$.

(a) (Markov) For $Y \geq 0$ and $t > 0$, prove $\mathbf{P}(Y \geq t) \leq \mathbf{E}Y/t$.

(b) (Chebyshev) Prove $\mathbf{P}(|X - \mu| \geq t) \leq \sigma^2/t^2$.

(c) (Cantelli) Prove the one-sided refinement $\mathbf{P}(X - \mu \geq t) \leq \frac{\sigma^2}{\sigma^2 + t^2}$ for $t > 0$.

Solution. *Idea.* Everything follows from one picture: a nonnegative variable that is sometimes $\geq t$ must, on average, be at least t times the chance of that. Chebyshev and Cantelli are Markov applied to well-chosen nonnegative functions of X .

(a) Since $Y \geq 0$, the pointwise inequality $t \mathbf{1}_{\{Y \geq t\}} \leq Y$ holds (on $\{Y \geq t\}$ it reads $t \leq Y$, elsewhere $0 \leq Y$). Taking expectations, $t \mathbf{P}(Y \geq t) \leq \mathbf{E}Y$.

(b) The event $\{|X - \mu| \geq t\}$ equals $\{(X - \mu)^2 \geq t^2\}$. Apply (a) to $Y = (X - \mu)^2$ at level t^2 :

$$\mathbf{P}(|X - \mu| \geq t) = \mathbf{P}((X - \mu)^2 \geq t^2) \leq \frac{\mathbf{E}(X - \mu)^2}{t^2} = \frac{\sigma^2}{t^2}.$$

So Chebyshev is just “Markov for the squared deviation.”

(c) For a one-sided bound we may *shift* before squaring, which is what gains the improvement. For any $u \geq 0$,

$$\{X - \mu \geq t\} \subseteq \{X - \mu + u \geq t + u\} \subseteq \{(X - \mu + u)^2 \geq (t + u)^2\},$$

so by (a), and using $\mathbf{E}(X - \mu + u)^2 = \sigma^2 + u^2$,

$$\mathbf{P}(X - \mu \geq t) \leq \frac{\sigma^2 + u^2}{(t + u)^2}.$$

This holds for every $u \geq 0$, so minimize the right side; the derivative vanishes at $u = \sigma^2/t$, giving the value $\sigma^2/(\sigma^2 + t^2)$. $\square$

Remark. The shift u is the whole trick: plain Chebyshev gives σ^2/t^2 , Cantelli the smaller $\sigma^2/(\sigma^2 + t^2) \leq 1$. All three decay only *polynomially* in t . Chebyshev already proves the weak law: for an average of n i.i.d. terms the variance is σ^2/n , so $\mathbf{P}(|\bar{X} - \mu| \geq t) \leq \sigma^2/(nt^2) \rightarrow 0$. The next problem does exponentially better when more moments are available.

Problem 2 (The Chernoff method). The single most important idea here. Chebyshev squares the deviation; the Chernoff method *exponentiates* it, punishing large values far more and turning polynomial tails into exponential ones — at the cost of needing the moment generating function $\mathbf{E}e^{\lambda X}$.

(a) Show that for any X and any t , $\mathbf{P}(X \geq t) \leq \inf_{\lambda > 0} e^{-\lambda t} \mathbf{E}e^{\lambda X}$.

(b) For $X \sim N(0, \sigma^2)$, deduce the Gaussian tail $\mathbf{P}(X \geq t) \leq e^{-t^2/(2\sigma^2)}$.

(c) For $S_n = \sum_{i=1}^n \varepsilon_i$ a sum of n independent signs, prove $\cosh \lambda \leq e^{\lambda^2/2}$ and conclude $\mathbf{P}(S_n \geq t) \leq e^{-t^2/(2n)}$.

Solution. *Idea.* For $\lambda > 0$, $x \mapsto e^{\lambda x}$ is increasing, so $\{X \geq t\}$ becomes $\{e^{\lambda X} \geq e^{\lambda t}\}$; Markov turns this into a bound through $\mathbf{E}e^{\lambda X}$, and we *optimize over* λ .

(a) For $\lambda > 0$, $\{X \geq t\} = \{e^{\lambda X} \geq e^{\lambda t}\}$, so Markov (Problem 1a) on $Y = e^{\lambda X} \geq 0$ gives $\mathbf{P}(X \geq t) \leq e^{-\lambda t} \mathbf{E}e^{\lambda X}$; take the infimum over $\lambda > 0$.

(b) By (F2), $\mathbf{E}e^{\lambda X} = e^{\sigma^2 \lambda^2/2}$, so we minimize $e^{-\lambda t + \sigma^2 \lambda^2/2}$. The parabolic exponent is least at $\lambda^* = t/\sigma^2$, where it equals $-t^2/(2\sigma^2)$. Hence $\mathbf{P}(X \geq t) \leq e^{-t^2/(2\sigma^2)}$ — the prototype every later bound is compared to.

(c) Using $(2k)! \geq 2^k k!$,

$$\cosh \lambda = \sum_{k \geq 0} \frac{\lambda^{2k}}{(2k)!} \leq \sum_{k \geq 0} \frac{\lambda^{2k}}{2^k k!} = \sum_{k \geq 0} \frac{(\lambda^2/2)^k}{k!} = e^{\lambda^2/2} :$$

a single sign has an MGF no larger than that of an $N(0, 1)$. By independence $\mathbf{E}e^{\lambda S_n} = (\cosh \lambda)^n \leq e^{n\lambda^2/2}$, and part (a) with $\lambda^* = t/n$ gives $\mathbf{P}(S_n \geq t) \leq e^{-t^2/(2n)}$. $\square$

Remark. The optimization in (a) is a Legendre transform of $\ln \mathbb{E}e^{\lambda X}$: the better you know the MGF, the sharper the tail. Part (c) is the precise sense in which a sum of bounded variables “behaves like a Gaussian of the same variance”: S_n has variance n and tail $e^{-t^2/(2n)}$, matching $N(0, n)$. By the central limit theorem $S_n/\sqrt{n} \Rightarrow N(0, 1)$; the comparison $\cosh \lambda \leq e^{\lambda^2/2}$ makes this non-asymptotic.

Problem 3 (Hoeffding’s lemma). To run the Chernoff method on a sum of bounded variables we need the MGF of *one* bounded mean-zero variable. Hoeffding’s lemma supplies it: such a variable has MGF no larger than that of a Gaussian with standard deviation $(b - a)/2$. Prove that if $\mathbb{E}X = 0$ and $a \leq X \leq b$ a.s., then $\mathbb{E}e^{\lambda X} \leq e^{\lambda^2(b-a)^2/8}$ for all λ .

Solution. *Idea.* The exponential is convex, so on $[a, b]$ it lies below the chord through its endpoints; taking expectations (with $\mathbb{E}X = 0$) replaces the random X by one deterministic function of λ , which a short Taylor estimate controls.

If $a = b$ then $X \equiv 0$ and both sides are 1; assume $a < b$. By convexity, for $x \in [a, b]$,

$$e^{\lambda x} \leq \frac{b - x}{b - a} e^{\lambda a} + \frac{x - a}{b - a} e^{\lambda b}.$$

Taking expectations and using $\mathbb{E}X = 0$ (the x -terms drop),

$$\mathbb{E}e^{\lambda X} \leq \frac{b e^{\lambda a} - a e^{\lambda b}}{b - a} =: e^{\varphi(h)}, \quad h := \lambda(b - a), \quad \theta := \frac{-a}{b - a} \in [0, 1],$$

where (using $\frac{b}{b-a} = 1-\theta$, $e^{\lambda a} = e^{-\theta h}$, $e^{\lambda b} = e^{(1-\theta)h}$) $\varphi(h) = -\theta h + \ln(1-\theta+\theta e^h)$. Here $\theta \in [0, 1]$ because $\mathbf{E}X = 0$ forces $a \leq 0 \leq b$. Now $\varphi(0) = 0$, $\varphi'(h) = -\theta + \frac{\theta e^h}{1-\theta+\theta e^h}$ gives $\varphi'(0) = 0$, and with $u(h) = \frac{\theta e^h}{1-\theta+\theta e^h} \in (0, 1)$,

$$\varphi''(h) = u(h)(1 - u(h)) \leq \frac{1}{4}.$$

By Taylor's theorem, $\varphi(h) = \frac{h^2}{2}\varphi''(\xi) \leq h^2/8$, so $\mathbf{E}e^{\lambda X} \leq e^{h^2/8} = e^{\lambda^2(b-a)^2/8}$. □

Remark. The “variance proxy” $(b-a)^2/4$ is the worst-case variance of a variable in $[a, b]$, attained by the symmetric two-point law on $\{a, b\}$ — a scaled Rademacher. So Hoeffding's lemma says: any bounded mean-zero variable is, MGF-wise, at most as spread out as the extreme sign variable.

Problem 4 (Hoeffding's inequality). Chernoff + Hoeffding's lemma over independent terms gives the workhorse tail for sums of bounded variables — behind countless confidence intervals and generalization bounds. Let $X_1, \dots, X_n$ be independent with $\mathbf{E}X_i = 0$, $a_i \leq X_i \leq b_i$.

(a) Prove $\mathbf{P}(|\sum_i X_i| \geq t) \leq 2 \exp(-2t^2 / \sum_i (b_i - a_i)^2)$.

(b) If the X_i are i.i.d. with mean μ and $a \leq X_i \leq b$, deduce $\mathbf{P}(|\bar{X} - \mu| \geq t) \leq 2e^{-2nt^2/(b-a)^2}$.

Solution. *Idea.* Chernoff turns the tail of a sum into a product of MGFs (independence); Hoeffding's lemma bounds each by a Gaussian MGF; then optimize λ as in the Gaussian case.

(a) For $\lambda > 0$, by Problem 2(a), independence, and Problem 3,

$$\mathbb{P}\left(\sum_i X_i \geq t\right) \leq e^{-\lambda t} \prod_{i=1}^n \mathbb{E} e^{\lambda X_i} \leq \exp\left(-\lambda t + \frac{\lambda^2}{8} \sum_i (b_i - a_i)^2\right).$$

The exponent is least at $\lambda^* = 4t / \sum_i (b_i - a_i)^2$, giving $-2t^2 / \sum_i (b_i - a_i)^2$. Applying this to $-X_1, \dots, -X_n$ as well and adding (union bound) gives the factor 2.

(b) Apply (a) to $Y_i = X_i - \mu \in [a - \mu, b - \mu]$, each of range $b - a$. Then $\sum_i Y_i = n(\bar{X} - \mu)$ and

$$\mathbb{P}(|n(\bar{X} - \mu)| \geq nt) \leq 2 \exp\left(-\frac{2(nt)^2}{n(b-a)^2}\right) = 2e^{-2nt^2/(b-a)^2}.$$

Remark. Compare with Chebyshev: for the i.i.d. mean it gave $\sigma^2/(nt^2)$ (decay $1/n$), whereas Hoeffding gives $e^{-2nt^2/(b-a)^2}$ (decay e^{-cn}). The price is boundedness; the reward is exponential confidence. Read backwards: $n \geq \frac{(b-a)^2}{2t^2} \ln \frac{2}{\alpha}$ samples give $|\bar{X} - \mu| \leq t$ with confidence $1 - \alpha$, for *any* distribution on $[a, b]$.

Problem 5 (Multiplicative Chernoff bound). For sums of indicators (counts), the yardstick is the *mean* μ , not the range, and one wants relative deviations $S \geq (1 + \delta)\mu$. The multiplicative Chernoff bound is sharp here — ubiquitous in randomized algorithms and balls-in-bins. Let $X_1, \dots, X_n$ be independent, $X_i \in \{0, 1\}$,

$P(X_i = 1) = p_i$, $S = \sum_i X_i$, $\mu = \mathbb{E}S$. Prove for $\delta > 0$,

$$P(S \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{1+\delta}} \right)^\mu \leq e^{-\mu\delta^2/3} \quad (0 < \delta \leq 1).$$

Solution. *Idea.* A single 0/1 variable has MGF $1 + p(e^\lambda - 1)$, which $1 + x \leq e^x$ folds into $e^{p(e^\lambda - 1)}$; the product over i sees the p_i only through μ , and the optimal λ is explicit.

For $\lambda > 0$, $\mathbb{E}e^{\lambda X_i} = 1 + p_i(e^\lambda - 1) \leq \exp(p_i(e^\lambda - 1))$ by (F1). By independence $\mathbb{E}e^{\lambda S} \leq \exp(\mu(e^\lambda - 1))$, so

$$P(S \geq (1 + \delta)\mu) \leq e^{-\lambda(1+\delta)\mu} \exp(\mu(e^\lambda - 1)).$$

At $\lambda = \ln(1 + \delta)$ the exponent is $\mu[\delta - (1 + \delta) \ln(1 + \delta)]$, giving the first bound. For $0 < \delta \leq 1$, the elementary $(1 + \delta) \ln(1 + \delta) \geq \delta + \delta^2/3$ yields $\leq e^{-\mu\delta^2/3}$. $\square$

Remark. Two regimes: for small δ the bound is Gaussian-looking, $e^{-\mu\delta^2/3}$ (relative fluctuations $\sim 1/\sqrt{\mu}$); for large δ the factor $(1 + \delta)^{-(1+\delta)}$ decays *super*-exponentially — this is why, with n balls in n bins, the fullest bin holds only $O(\log n / \log \log n)$ balls w.h.p.

Problem 6 (Bennett's and Bernstein's inequalities). Hoeffding ignores variance and is wasteful when variances are small. Bennett/Bernstein use the variance, giving a Gaussian tail at the variance scale for moderate t and an exponential tail for large t . Let $X_1, \dots, X_n$ be independent, $\mathbb{E}X_i = 0$, $X_i \leq b$ a.s., $\sigma^2 = \sum_i \text{var}(X_i)$.

- (a) (Bennett) Prove $\mathbf{P}(\sum_i X_i \geq t) \leq \exp(-\frac{\sigma^2}{b^2} h(\frac{bt}{\sigma^2}))$, $h(u) = (1+u) \ln(1+u) - u$.
 (b) (Bernstein) Using $h(u) \geq \frac{u^2}{2(1+u/3)}$, deduce $\mathbf{P}(\sum_i X_i \geq t) \leq \exp(-\frac{t^2}{2(\sigma^2+bt/3)})$.

Solution. *Idea.* The one new ingredient is a variance-aware MGF bound: write $e^{\lambda x} - 1 - \lambda x = (\lambda x)^2 \varphi(\lambda x)$ and use that φ increases ((F5)) to replace $\varphi(\lambda x)$ by $\varphi(\lambda b)$; the leftover x^2 becomes the variance.

(a) Fix $\lambda > 0$. For $x \leq b$, since φ is nondecreasing,

$$e^{\lambda x} - 1 - \lambda x = (\lambda x)^2 \varphi(\lambda x) \leq (\lambda x)^2 \varphi(\lambda b) = \frac{x^2}{b^2} (e^{\lambda b} - 1 - \lambda b).$$

Take expectations (the linear term drops as $\mathbf{E}X_i = 0$) and use $1 + y \leq e^y$ ((F1)):

$$\mathbf{E}e^{\lambda X_i} \leq 1 + \frac{\text{var}(X_i)}{b^2} (e^{\lambda b} - 1 - \lambda b) \leq \exp\left(\frac{\text{var}(X_i)}{b^2} (e^{\lambda b} - 1 - \lambda b)\right).$$

Multiplying over i and running Chernoff, $\mathbf{P}(\sum_i X_i \geq t) \leq \exp(-\lambda t + \frac{\sigma^2}{b^2} (e^{\lambda b} - 1 - \lambda b))$; minimizing at $\lambda = \frac{1}{b} \ln(1 + \frac{bt}{\sigma^2})$ gives $-\frac{\sigma^2}{b^2} h(\frac{bt}{\sigma^2})$.

(b) Substitute $h(u) \geq \frac{u^2}{2(1+u/3)}$ with $u = bt/\sigma^2$: the exponent is at most $-\frac{\sigma^2}{b^2} \cdot \frac{(bt/\sigma^2)^2}{2(1+bt/(3\sigma^2))} = -\frac{t^2}{2(\sigma^2+bt/3)}$. $\square$

Remark. Read Bernstein in its two regimes. For $t \lesssim \sigma^2/b$ the denominator is $\approx 2\sigma^2$, giving the Gaussian tail $e^{-t^2/(2\sigma^2)}$ set by the variance; for $t \gtrsim \sigma^2/b$ the $bt/3$ term dominates, giving the exponential tail $e^{-3t/(2b)}$ set

by the bound b . Whenever the variance is far below the range, Bernstein beats Hoeffding (which only sees $\sum(b_i - a_i)^2$).

From sums to maxima. The bounds so far control one sum; often we must control many quantities at once — the largest of N statistics. The next problem shows the cost is only logarithmic.

Problem 7 (Maximal inequality for Gaussian-type variables). Let $X_1, \dots, X_N$ (*not* necessarily independent) each satisfy $\mathbb{E}e^{\lambda X_i} \leq e^{\sigma^2 \lambda^2 / 2}$ for all λ — e.g. each $N(0, \sigma^2)$, or (by Problem 3) bounded mean-zero.

(a) Show $\mathbb{P}(\max_{i \leq N} X_i \geq t) \leq N e^{-t^2/(2\sigma^2)}$.

(b) Show $\mathbb{E}[\max_{i \leq N} X_i] \leq \sigma \sqrt{2 \ln N}$.

Solution. *Idea.* For the tail, a max exceeds t only if some term does (union bound + Chernoff, no independence needed). For the expectation, exponentiate the max, push expectation onto the sum, and optimize.

(a) $\{\max_i X_i \geq t\} = \bigcup_i \{X_i \geq t\}$, and by Problem 2(a) with the hypothesis, $\mathbb{P}(X_i \geq t) \leq \inf_{\lambda > 0} e^{-\lambda t + \sigma^2 \lambda^2 / 2} = e^{-t^2/(2\sigma^2)}$. The union bound ((F6)) gives $\mathbb{P}(\max_i X_i \geq t) \leq N e^{-t^2/(2\sigma^2)}$.

(b) For $\lambda > 0$, Jensen ((F1)) and then bounding the max by the sum give

$$e^{\lambda \mathbb{E}[\max_i X_i]} \leq \mathbb{E}e^{\lambda \max_i X_i} = \mathbb{E} \max_i e^{\lambda X_i} \leq \sum_i \mathbb{E}e^{\lambda X_i} \leq N e^{\sigma^2 \lambda^2 / 2}.$$

Hence $\mathbb{E}[\max_i X_i] \leq \frac{\ln N}{\lambda} + \frac{\sigma^2 \lambda}{2}$; optimizing at $\lambda = \sqrt{2 \ln N} / \sigma$ gives $\sigma \sqrt{2 \ln N}$. □

Remark. The maximum of N Gaussian-type variables sits only $\sqrt{2 \ln N}$ standard deviations out — logarithmic growth, sharp for independent Gaussians. This $\sqrt{\log N}$ is why union-bound arguments over finite classes work so well, and the seed of chaining bounds for suprema.

Problem 8 (The Paley–Zygmund (second-moment) inequality). Every bound so far is an *upper* bound on a tail. Sometimes we need the opposite: a guarantee that X is large with non-negligible probability (anti-concentration). The second-moment method delivers this from two moments. Let $X \geq 0$ with $0 < \mathbf{E}X^2 < \infty$; prove for $\theta \in [0, 1]$,

$$\mathbf{P}(X > \theta \mathbf{E}X) \geq (1 - \theta)^2 \frac{(\mathbf{E}X)^2}{\mathbf{E}X^2}.$$

Solution. *Idea.* The mean cannot all come from the event $\{X \leq \theta \mathbf{E}X\}$ (that contributes $\leq \theta \mathbf{E}X$), so a definite amount comes from $\{X > \theta \mathbf{E}X\}$; Cauchy–Schwarz turns that contribution into a probability. Split $\mathbf{E}X = \mathbf{E}[X \mathbf{1}_{\{X \leq \theta \mathbf{E}X\}}] + \mathbf{E}[X \mathbf{1}_{\{X > \theta \mathbf{E}X\}}]$; the first term is $\leq \theta \mathbf{E}X$. For the second, Cauchy–Schwarz ((F1)) with the indicator gives

$$\mathbf{E}[X \mathbf{1}_{\{X > \theta \mathbf{E}X\}}] \leq (\mathbf{E}X^2)^{1/2} \mathbf{P}(X > \theta \mathbf{E}X)^{1/2}.$$

Hence $(1 - \theta)\mathbf{E}X \leq (\mathbf{E}X^2)^{1/2} \mathbf{P}(X > \theta \mathbf{E}X)^{1/2}$; squaring and rearranging gives the claim. $\square$

Remark. The ratio $(\mathbf{E}X)^2/\mathbf{E}X^2 \in (0, 1]$ measures how concentrated X is. This is the *second-moment method*: to show a nonnegative count (e.g. the number of triangles in a random graph) is positive w.h.p., show its mean is large and its second moment is comparable to the mean squared.

Problem 9 (Sharpness for Rademacher sums). Is Hoeffding's exponent tight? For Rademacher sums we can compute the MGF exactly and see that the Gaussian comparison of Problem 2(c) is the best possible, so the exponent $t^2/(2n)$ cannot be improved by this method. Let $S_n = \sum_{i=1}^n \varepsilon_i$.

(a) Compute the exact MGF $\mathbb{E}e^{\lambda S_n}$.

(b) Show $\cosh \lambda \leq e^{\lambda^2/2}$ is tight to second order at $\lambda = 0$, so the exponent $t^2/(2n)$ cannot be replaced by ct^2/n with $c < \frac{1}{2}$.

Solution. (a) By independence and $\mathbb{E}e^{\lambda \varepsilon_i} = \frac{1}{2}(e^\lambda + e^{-\lambda}) = \cosh \lambda$, we get $\mathbb{E}e^{\lambda S_n} = (\cosh \lambda)^n$ — no inequality.

(b) Compare series: $\cosh \lambda = 1 + \frac{\lambda^2}{2} + \frac{\lambda^4}{24} + \dots$ and $e^{\lambda^2/2} = 1 + \frac{\lambda^2}{2} + \frac{\lambda^4}{8} + \dots$ agree through λ^2 , so $\ln \cosh \lambda = \frac{\lambda^2}{2}(1 + o(1))$ as $\lambda \rightarrow 0$. The Chernoff exponent is $\sup_\lambda \{\lambda t - n \ln \cosh \lambda\}$; a bound $\ln \cosh \lambda \leq c\lambda^2$ with $c < \frac{1}{2}$ fails for small λ , so no exponent beats the value at $c = \frac{1}{2}$, namely $e^{-t^2/(2n)}$. Hoeffding is optimal at this level. $\square$

Remark. This matches the central limit theorem: $S_n/\sqrt{n} \Rightarrow N(0, 1)$, tail $e^{-t^2/2}$, i.e. $\mathbb{P}(S_n \geq t) \approx e^{-t^2/(2n)}$ for $t = O(\sqrt{n})$. The MGF comparison turns this asymptotic truth into a bound valid for every n, t .

Squared Gaussians. We leave sums of bounded variables for sums of *squares* of Gaussians — the χ^2 distribution — whose MGF is again explicit, so the Chernoff method applies verbatim.

Problem 10 (χ^2 concentration). The $\chi_n^2 = \sum_i Z_i^2$ law governs squared lengths of Gaussian vectors, hence variance estimates, goodness-of-fit, and random projections. Let $Z_1, \dots, Z_n$ be i.i.d. $N(0, 1)$.

- (a) Show $\mathbb{E}e^{\mu\chi_n^2} = (1 - 2\mu)^{-n/2}$ for $\mu < \frac{1}{2}$.
 (b) Prove $\mathbb{P}(\chi_n^2 \geq n(1 + \delta)) \leq e^{-n\delta^2/8}$ for $\delta \in (0, 1]$.

Solution. *Idea.* A single Z^2 has explicit MGF $(1 - 2\mu)^{-1/2}$; raising to the n th power and running Chernoff reduces everything to choosing μ well.

(a) By (F2), $\mathbb{E}e^{\mu Z_i^2} = (1 - 2\mu)^{-1/2}$ for $\mu < \frac{1}{2}$; independence gives $\mathbb{E}e^{\mu\chi_n^2} = (1 - 2\mu)^{-n/2}$.

(b) Chernoff gives $\mathbb{P}(\chi_n^2 \geq n(1 + \delta)) \leq e^{-\mu n(1 + \delta)}(1 - 2\mu)^{-n/2}$ for $0 < \mu < \frac{1}{2}$. The neat choice $\mu = \frac{\delta}{2(1 + \delta)}$ makes $1 - 2\mu = \frac{1}{1 + \delta}$ and collapses the exponent:

$$-\frac{n\delta}{2(1 + \delta)}(1 + \delta) + \frac{n}{2} \ln(1 + \delta) = -\frac{n}{2}(\delta - \ln(1 + \delta)) \leq -\frac{n}{2} \cdot \frac{\delta^2}{4} = -\frac{n\delta^2}{8},$$

using $\delta - \ln(1 + \delta) \geq \delta^2/4$ on $[0, 1]$. □

Remark. So $\chi_n^2/n \rightarrow 1$ with fluctuations of order $1/\sqrt{n}$: a length- n Gaussian vector has squared norm very close to n . By diagonalization ((F4)), the general quadratic form $\mathbf{g}'\mathbf{A}\mathbf{g}$ has MGF $\prod_k (1 - 2\mu\lambda_k)^{-1/2}$, so the same method handles weighted sums of squares. With the lower tail (next problem) this powers Johnson–Lindenstrauss.

Problem 11 (Lower-tail Chernoff bounds). Our tails have all been upper tails; the lower tails complete the picture and come from the same method with a negative parameter ($e^{-\lambda X}$).

- (a) For S as in Problem 5 and $0 < \delta < 1$, prove $\mathbf{P}(S \leq (1 - \delta)\mu) \leq e^{-\mu\delta^2/2}$, where $\mu = \mathbf{E}S$.
- (b) For $\chi_n^2 = \sum_i Z_i^2$ and $\delta \in (0, 1)$, prove $\mathbf{P}(\chi_n^2 \leq n(1 - \delta)) \leq e^{-n\delta^2/4}$, hence $\mathbf{P}(|\chi_n^2 - n| \geq n\delta) \leq 2e^{-n\delta^2/8}$.

Solution. *Idea.* A lower tail $\{S \leq s\}$ is $\{e^{-\lambda S} \geq e^{-\lambda s}\}$ for $\lambda > 0$; apply Markov to $e^{-\lambda S}$ and optimize.

(a) For $\lambda > 0$, $\mathbf{P}(S \leq (1 - \delta)\mu) \leq e^{\lambda(1 - \delta)\mu} \mathbf{E}e^{-\lambda S}$, and (as in Problem 5, now with $-\lambda$) $\mathbf{E}e^{-\lambda S} \leq \exp(\mu(e^{-\lambda} - 1))$. So the bound is $\exp(\mu[\lambda(1 - \delta) + e^{-\lambda} - 1])$; at $\lambda = -\ln(1 - \delta)$ the exponent is $\mu[-\delta - (1 - \delta)\ln(1 - \delta)] \leq -\mu\delta^2/2$ (using $(1 - \delta)\ln(1 - \delta) \geq -\delta + \delta^2/2$).

(b) For $s > 0$, $\mathbf{P}(\chi_n^2 \leq n(1 - \delta)) \leq e^{sn(1 - \delta)}(1 + 2s)^{-n/2}$ by Problem 10(a) with $\mu = -s$. At $1 + 2s = (1 - \delta)^{-1}$ the exponent is $\frac{n}{2}(\delta + \ln(1 - \delta)) \leq -\frac{n\delta^2}{4}$, since $\ln(1 - \delta) \leq -\delta - \delta^2/2$. Adding the upper tail of Problem 10(b), $\mathbf{P}(|\chi_n^2 - n| \geq n\delta) \leq e^{-n\delta^2/8} + e^{-n\delta^2/4} \leq 2e^{-n\delta^2/8}$. $\square$

Remark. Lower tails are often slightly stronger (here $e^{-n\delta^2/4}$ vs $e^{-n\delta^2/8}$): a sum of nonnegative pieces finds it harder to fall far below its mean than to rise far above. The two-sided χ^2 bound is exactly what the next problem needs.

Problem 12 (The Johnson–Lindenstrauss lemma). The payoff. Any N points in a high-dimensional space can be linearly mapped into dimension $m \approx \varepsilon^{-2} \ln N$ — independent of the original dimension — while distorting all pairwise distances by at most $1 \pm \varepsilon$, via pure χ^2 concentration and a union bound. Let $x_1, \dots, x_N \in \mathbb{R}^D$, $\varepsilon \in (0, 1)$; let $\mathbf{A}$ be $m \times D$ with i.i.d. $N(0, 1)$ entries and $f(x) = \mathbf{A}x/\sqrt{m}$. Show that if

$m \geq 24\varepsilon^{-2} \ln N$ then, with positive probability,

$$(1 - \varepsilon)\|x_i - x_j\|^2 \leq \|f(x_i) - f(x_j)\|^2 \leq (1 + \varepsilon)\|x_i - x_j\|^2 \quad \text{for all } i, j.$$

Solution. *Idea.* A Gaussian projection of a *fixed* vector has squared length exactly $\frac{1}{m}\chi_m^2$ times the true squared length; concentration keeps this within $1 \pm \varepsilon$, and a union bound covers all pairs.

Fix $i \neq j$ and set $u = x_i - x_j \neq 0$. The rows $r_1, \dots, r_m$ of $\mathbf{A}$ are i.i.d. $N(\mathbf{0}, \mathbf{I}_D)$, so $\langle r_k, u \rangle / \|u\|$ are i.i.d. $N(0, 1)$ and

$$\frac{\|f(x_i) - f(x_j)\|^2}{\|u\|^2} = \frac{1}{m} \sum_{k=1}^m \left(\frac{\langle r_k, u \rangle}{\|u\|} \right)^2 = \frac{\chi_m^2}{m}.$$

By the two-sided bound of Problem 11(b), this single pair is distorted by more than $1 \pm \varepsilon$ with probability $\leq 2e^{-m\varepsilon^2/8}$. There are fewer than N^2 pairs, so by the union bound ((F6)) the chance that *some* pair fails is $\leq N^2 \cdot 2e^{-m\varepsilon^2/8}$. When $m \geq 24\varepsilon^{-2} \ln N$ we have $m\varepsilon^2/8 \geq 3 \ln N$, so this is $\leq 2N^2 e^{-3 \ln N} = 2/N < 1$ for $N \geq 3$. A failure probability below 1 means some $\mathbf{A}$ works for all pairs at once. $\square$

Remark. The target dimension $m \approx \varepsilon^{-2} \ln N$ depends only on the number of points and the accuracy, never on the ambient dimension D — the striking content of the lemma. The projection is also *oblivious* (drawn without seeing the data), which is why random projections are routine in nearest-neighbor search, clustering, and sketching.

A different kind of inequality. The remaining problems concern not concentration but *positive correlation*: for

a centered Gaussian, two symmetric events help rather than hinder one another, $\mu(K \cap L) \geq \mu(K)\mu(L)$.

Problem 13 (Gaussian measure of a disk; two easy instances of the GCI). Two warm-ups where “ $\mu(K \cap L) \geq \mu(K)\mu(L)$ ” is transparent — to fix what the inequality claims and see its two extremes. Let μ be a standard Gaussian measure.

- (a) For $G \sim N(\mathbf{0}, \mathbf{I}_2)$, show $\mu(\|x\| \leq R) = 1 - e^{-R^2/2}$.
- (b) (Nested) For $K = \{\|x\| \leq R_1\} \subseteq L = \{\|x\| \leq R_2\}$, show $\mu(K \cap L) \geq \mu(K)\mu(L)$ directly; evaluate at $R_1^2 = 2 \ln 2$, $R_2^2 = 2 \ln 4$.
- (c) (Independent blocks) For $N(\mathbf{0}, \mathbf{I}_3)$, $K = \{|x_1| \leq a\}$, $L = \{x_2^2 + x_3^2 \leq R^2\}$, show $\mu(K \cap L) = \mu(K)\mu(L)$.

Solution. (a) In polar coordinates ($N(\mathbf{0}, \mathbf{I}_2)$ density $\frac{1}{2\pi}e^{-r^2/2}$, area element $r \, dr \, d\theta$),

$$\mu(\|x\| \leq R) = \int_0^{2\pi} \int_0^R \frac{1}{2\pi} e^{-r^2/2} r \, dr \, d\theta = \int_0^R r e^{-r^2/2} dr = 1 - e^{-R^2/2}.$$

(b) Here $K \subseteq L$, so $K \cap L = K$ and $\mu(K \cap L) = \mu(K) \geq \mu(K)\mu(L)$ because $\mu(L) \leq 1$. By (a), $\mu(K) = 1 - e^{-\ln 2} = \frac{1}{2}$, $\mu(L) = 1 - e^{-2 \ln 2} = \frac{3}{4}$, and $\frac{1}{2} \geq \frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8}$.

(c) The density factors across $\{1\}$ and $\{2, 3\}$ as $\frac{e^{-x_1^2/2}}{\sqrt{2\pi}} \cdot \frac{e^{-(x_2^2+x_3^2)/2}}{2\pi}$, and $K \cap L$ is the product of the slab and the disk on those blocks. By Fubini, $\mu(K \cap L) = \mu(K)\mu(L)$, an exact equality ($\mu(K) = \text{erf}(a/\sqrt{2})$, $\mu(L) = 1 - e^{-R^2/2}$). $\square$

Remark. These are the two endpoints of the GCI: very slack when one set sits inside the other (b), and an exact equality when the sets constrain *independent* coordinates (c). The content lies between — sets constraining overlapping, correlated directions, the subject of the next problem.

Problem 14 (The bivariate Gaussian correlation inequality, via Plackett). The Gaussian correlation inequality (conjectured in the 1950s, proved in general by Royen in 2014) is hard in high dimensions, but its first nontrivial case — two correlated symmetric slabs in the plane — is completely elementary via Plackett's formula for how a bivariate normal probability changes with ρ . Let $X \sim N(\mathbf{0}, \mathbf{C}_\rho)$, $\mathbf{C}_\rho = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$, $|\rho| < 1$, $K = \{|x_1| \leq a\}$, $L = \{|x_2| \leq a\}$; Φ_2, ϕ_2 the bivariate normal CDF/density.

(a) (Plackett) Prove $\partial_\rho \Phi_2(h, k; \rho) = \phi_2(h, k; \rho)$.

(b) With $P(\rho) = \mu(K \cap L)$, prove $P'(\rho) = \frac{1}{\pi \sqrt{1 - \rho^2}} (e^{-a^2/(1+\rho)} - e^{-a^2/(1-\rho)})$.

(c) Conclude $\mu(K \cap L) > \mu(K)\mu(L)$ for every $\rho \neq 0$, with equality iff $\rho = 0$.

Solution. *Idea.* Differentiating the box probability in ρ reduces, via Plackett, to the density at the four corners; their exponents are explicit, so the derivative has a fixed sign, and integrating from the independent case $\rho = 0$ gives strict inequality.

(a) Write the density as the inverse Fourier transform of the characteristic function ((F3)),

$$\phi_2(h, k; \rho) = \frac{1}{(2\pi)^2} \iint e^{-i(sh+tk)} e^{-\frac{1}{2}(s^2+2\rho st+t^2)} ds dt.$$

Differentiating under the integral, ∂_ρ brings down $-st$ and $\partial_h \partial_k$ brings down $(-is)(-it) = -st$, so $\partial_\rho \phi_2 = \partial^2 \phi_2 / \partial h \partial k$. Integrating over $(-\infty, h] \times (-\infty, k]$ (with ϕ_2 and its first partials vanishing at $-\infty$), two uses of the fundamental theorem of calculus give $\partial_\rho \Phi_2 = \phi_2$.

(b) By inclusion–exclusion, $P(\rho) = \Phi_2(a, a) - \Phi_2(-a, a) - \Phi_2(a, -a) + \Phi_2(-a, -a)$; differentiate and apply (a). The exponent $Q(h, k) = \frac{h^2 - 2\rho hk + k^2}{2(1-\rho^2)}$ gives $Q(\pm a, \pm a) = \frac{a^2}{1+\rho}$ and $Q(\pm a, \mp a) = \frac{a^2}{1-\rho}$ (each twice).

With prefactor $\frac{1}{2\pi\sqrt{1-\rho^2}}$, the like-sign corners contribute $e^{-a^2/(1+\rho)}$ and the opposite-sign corners $e^{-a^2/(1-\rho)}$, so $P'(\rho) = \frac{1}{\pi\sqrt{1-\rho^2}}(e^{-a^2/(1+\rho)} - e^{-a^2/(1-\rho)})$.

(c) For $\rho \in (0, 1)$, $0 < 1-\rho < 1+\rho$ gives $\frac{a^2}{1+\rho} < \frac{a^2}{1-\rho}$, hence $P'(\rho) > 0$. The reflection $(x_1, x_2) \mapsto (x_1, -x_2)$ sends $\mathbf{C}_\rho \rightarrow \mathbf{C}_{-\rho}$ and fixes the box, so P is even; thus P increases on $[0, 1)$ from $P(0) = (2\Phi(a) - 1)^2 = \mu(K)\mu(L)$. Hence $\mu(K \cap L) > \mu(K)\mu(L)$ for every $\rho \neq 0$. $\square$

Remark. The inequality says positively (or negatively) correlated symmetric slabs make each other *more* likely than independence predicts. In statistics this is the basis of the Šidák correction: simultaneous confidence intervals for correlated Gaussian estimates have at least the coverage one would get pretending they were independent.

Problem 15 (How slack is the inequality? Small boxes). The inequality is strict, but by how much? For small boxes the answer is a clean closed form measuring the “correlation gain.” In the setting of Problem 14,

show that for fixed $\rho \neq 0$,

$$\frac{\mu(K \cap L)}{\mu(K)\mu(L)} \xrightarrow{a \rightarrow 0} \frac{1}{\sqrt{1 - \rho^2}} > 1.$$

Solution. *Idea.* Both the gap $\mu(K \cap L) - \mu(K)\mu(L)$ and the product $\mu(K)\mu(L)$ are of order a^2 ; computing each leading coefficient leaves a constant.

By Problem 14, $\mu(K \cap L) - \mu(K)\mu(L) = \int_0^{|\rho|} P'(s) ds$. Expanding $e^{-a^2/(1 \pm s)} = 1 - \frac{a^2}{1 \pm s} + O(a^4)$,

$$e^{-a^2/(1+s)} - e^{-a^2/(1-s)} = a^2 \left(\frac{1}{1-s} - \frac{1}{1+s} \right) + O(a^4) = \frac{2a^2 s}{1-s^2} + O(a^4),$$

so $\int_0^{|\rho|} \frac{2a^2 s}{\pi(1-s^2)^{3/2}} ds = \frac{2a^2}{\pi} ((1-\rho^2)^{-1/2} - 1) + O(a^4)$. Since $\text{erf}(a/\sqrt{2}) = \sqrt{2/\pi} a + O(a^3)$ gives $\mu(K)\mu(L) = (2\Phi(a) - 1)^2 = \frac{2a^2}{\pi} + O(a^4)$, adding it back and dividing,

$$\frac{\mu(K \cap L)}{\mu(K)\mu(L)} = \frac{\frac{2a^2}{\pi}(1-\rho^2)^{-1/2} + O(a^4)}{\frac{2a^2}{\pi} + O(a^4)} \xrightarrow{a \rightarrow 0} (1-\rho^2)^{-1/2} > 1.$$

Remark. The gain $(1-\rho^2)^{-1/2}$ is 1 at $\rho = 0$ (independence, no gain) and blows up as $\rho \rightarrow \pm 1$ (near-perfect dependence, where the two events nearly coincide). It quantifies how conservative it is to treat correlated tests as independent — the more correlated, the more the true joint coverage exceeds the independence value.